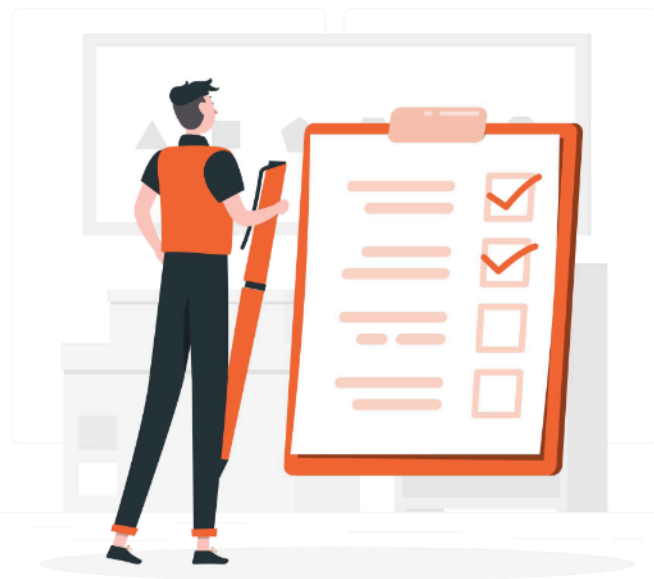




Prompt Optimization

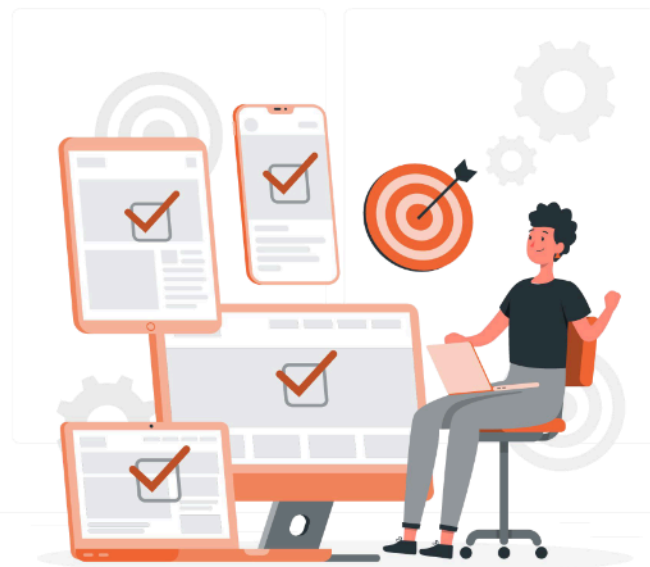
Lesson

1. Chain-of-Thought prompting
2. Self-check prompting
3. Retrieval-aware prompting
4. Temperature, top-p, top-k tuning
5. Managing creativity vs accuracy
6. Context Window Management
7. Cost optimization techniques



Learning Outcome

1. Master advanced prompting to improve AI reasoning and accuracy.
2. Optimize model behavior for balanced creativity and precision.
3. Manage tokens and prompts efficiently within limits and budget.





Why We Need Prompt Optimization?

Prompting Problem

Although AI adoption is massive across various lines of work, many users get trapped in inefficient “prompting loops.”

Without optimization:

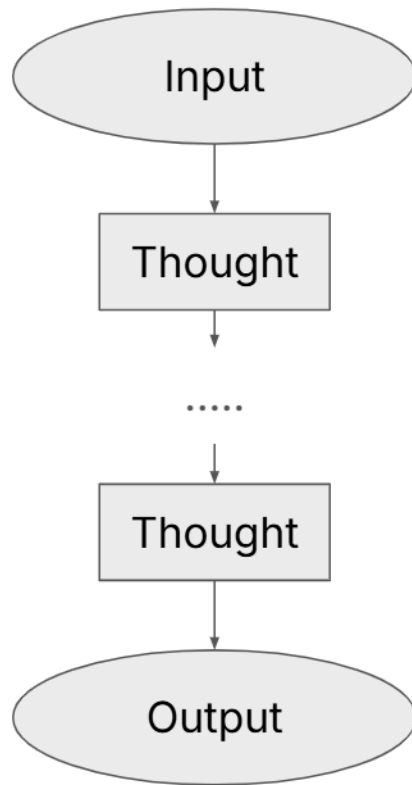
- **Hallucination:** Outputs that are not aligned with facts.
- **Wasted Initiative:** Repetition without clear results.
- **Token Exhaustion:** Reaching the limit without a solution.

Chain-of- Thought Prompting



Chain-of-Thought Definition

Prompting technique to generate **step-by-step reasoning** before giving the final answer.



Example of Chain-of-Thought

Bandingkan kedua proyek ini

Proyek A membutuhkan investasi 2 miliar rupiah dengan estimasi ROI 15% per tahun.

Proyek B membutuhkan investasi 1,2 miliar rupiah dengan estimasi ROI 10% per tahun tetapi memiliki risiko lebih rendah.

Without CoT: it usually gives general suggestions directly.

Berikut perbandingan Proyek A dan Proyek B secara finansial dan risiko:

1 Investasi Awal

- Proyek A: Rp 2.000.000.000
- Proyek B: Rp 1.200.000.000

2 Proyek B membutuhkan modal lebih kecil (lebih ringan dari sisi cash flow dan pendanaan).

2 Estimasi Return per Tahun

- Proyek A: $15\% \times 2 \text{ M} = \text{Rp } 300.000.000$ per tahun
- Proyek B: $10\% \times 1,2 \text{ M} = \text{Rp } 120.000.000$ per tahun

3 Secara nominal, Proyek A menghasilkan laba tahunan lebih besar.

3 Persentase Pengembalian (ROI)

- Proyek A: 15%
- Proyek B: 10%

4 Secara persentase, Proyek A lebih menguntungkan.

4 Risiko

- Proyek A: Risiko lebih tinggi (diasumsikan karena ROI lebih tinggi).
- Proyek B: Risiko lebih rendah (lebih stabil).

5 Jika Anda konservatif atau mengutamakan stabilitas, Proyek B lebih aman.

5 Analisis Strategis

Faktor	Proyek A	Proyek B
Modal Awal	Lebih besar	Lebih kecil
Laba Tahunan	Lebih tinggi	Lebih rendah
ROI	Lebih tinggi	Lebih rendah
Risiko	Lebih tinggi	Lebih rendah
Cocok untuk	Investor agresif	Investor konservatif

6 Kesimpulan

- Pilih Proyek A jika:
 - Anda siap dengan risiko lebih tinggi
 - Mengejar return maksimal
 - Memiliki kapasitas modal cukup
- Pilih Proyek B jika:
 - Ingin risiko lebih rendah
 - Ingin menjaga stabilitas arus kas
 - Modal terbatas

Example of Chain-of-Thought

Perusahaan mempertimbangkan dua proyek:

Proyek A membutuhkan investasi 2 miliar rupiah dengan estimasi ROI 15% per tahun.

Proyek B membutuhkan investasi 1,2 miliar rupiah dengan estimasi ROI 10% per tahun tetapi memiliki risiko lebih rendah.

Evaluasi kedua opsi tersebut dan jelaskan langkah analisis yang kamu gunakan sebelum memberikan rekomendasi.

With CoT: the response is more **structured and logical**.

Untuk mengevaluasi Proyek A dan Proyek B, saya menggunakan beberapa langkah analisis keuangan dasar berikut:

Langkah Analisis

1 Menghitung Estimasi Return Tahunan (Nominal)

Rumus sederhana:

$$\text{Return} = \text{Investasi} \times \text{ROI}$$

Proyek A

- Investasi = Rp 2.000.000.000
- ROI = 15%
- Return tahunan = $2.000.000.000 \times 15\% = \text{Rp } 300.000.000$

Proyek B

- Investasi = Rp 1.200.000.000
- ROI = 10%
- Return tahunan = $1.200.000.000 \times 10\% = \text{Rp } 120.000.000$

Secara nominal, Proyek A menghasilkan keuntungan tahunan lebih besar.

2 Menghitung Efisiensi Modal (Return per Rupiah)

Karena ROI sudah dalam bentuk persentase, kita lihat efisiensi penggunaan modal:

- Proyek A: 15%
- Proyek B: 10%

Proyek A lebih efisien dalam menghasilkan return dari setiap rupiah yang diinvestasikan.

Analisa Risiko

Diketahui:

- Proyek A → Return lebih tinggi, risiko lebih tinggi (implisit)
- Proyek B → Return lebih rendah, risiko lebih rendah

Dari teori keuangan, ada prinsip **risk-return tradeoff**:

Semakin tinggi potensi return, biasanya semakin tinggi risiko.

Jika perusahaan:

- Agresif / growth-oriented** → cenderung memilih Proyek A
- Konservatif / menjaga stabilitas arus kas** → cenderung memilih Proyek B

Pertimbangan Skala Investasi & Likuiditas

Proyek A membutuhkan tambahan modal Rp 800 juta dibanding Proyek B.

Perlu dianalisis:

- Apakah perusahaan memiliki dana cukup?
- Apakah ada proyek alternatif lain untuk sisa Rp 800 juta jika memilih Proyek B?
- Dampak terhadap arus kas dan leverage perusahaan?

Risk-Adjusted Return (Pendekatan Konseptual)

Idealnya, kita menghitung:

- NPV (Net Present Value)
- IRR (Internal Rate of Return)
- Payback Period
- Analisis sensitivitas

Namun karena data terbatas (tidak ada horizon waktu & discount rate), kita hanya bisa evaluasi berdasarkan ROI dan profil risiko.

Ringkasan Perbandingan

Aspek	Proyek A	Proyek B
Investasi	2 M	1,2 M
ROI	15%	10%
Return Tahunan	300 juta	120 juta
Risiko	Lebih tinggi	Lebih rendah
Kebutuhan Modal	Lebih besar	Lebih kecil

Rekomendasi

✓ Jika perusahaan memiliki:

- Toleransi risiko tinggi
- Modal cukup
- Target pertumbuhan agresif

→ **Pilih Proyek A**

✓ Jika perusahaan:

- Mengutamakan stabilitas
- Ingin menjaga arus kas
- Memiliki keterbatasan modal

→ **Pilih Proyek B**

Kesimpulan Akhir

Secara finansial murni (tanpa mempertimbangkan risiko detail), Proyek A lebih menguntungkan karena ROI dan return absolut lebih tinggi.

Namun keputusan akhir sangat bergantung pada:

- Profil risiko perusahaan
- Ketersediaan modal
- Strategi jangka panjang perusahaan

Example of CoT Sentences

General Phrases

1. Explain step by step before providing the final answer.
2. Break down your reasoning process systematically.
3. Show your reasoning process.
4. Analyze this problem in a structured way before answering.

More Professional

1. Conduct a step-by-step analysis before providing recommendations.
2. Identify the key variables first, then explain the relationships between them.
3. Build the argument logically from premise to conclusion.
4. Present risk considerations and trade-offs before determining the final decision.

Analytical Version

1. Formulate the hypothesis first, then evaluate it systematically.
2. Explain the analytical framework used in this analysis.
3. Detail the inferences made at each stage.
4. Include logical justification for every step.

Self-Check Prompting



Self-Check Definition

Prompting technique where the AI is instructed to **review, critique, and correct its own initial output** before finalizing the response.



Example & Structure of Self-Check

1

Initial Answer

1 Identifikasi Risiko Utama & Strategi Mitigasi

A. Risiko Strategis & Manajemen Proyek

1. Scope Creep (Perluasan ruang lingkup)

Risiko: Pemintaan tambahan fitur di tengah proyek menyebabkan keterlambatan dan pembengkakan biaya.

Mitigasi:

- Tetapkan scope baseline yang disetujui manajemen.
- Gunakan change control board (CCB).
- Prioritaskan kebutuhan: must-have vs nice-to-have.

2. Timeline Tidak Realistik (6 bulan terlalu agresif)

Risiko: Sistem belum stabil saat go-live.

Mitigasi:

- Lakukan gap analysis awal yang komprehensif.
- Gunakan pendekatan phased rollout jika memungkinkan.
- Siapkan buffer untuk testing & training.

3. Kurangnya Dukungan Manajemen Puncak

Risiko: Minimnya komitmen lintas departemen.

Mitigasi:

- Bentuk steering committee.
- Lakukan komunikasi rutin progress ke direksi.
- Tetapkan sponsor eksekutif.

Sebuah perusahaan akan mengimplementasikan sistem ERP baru dalam 6 bulan.

- Identifikasi risiko utama dan strategi mitigasinya.
- Tinjau kembali daftar risiko tersebut dan periksa apakah ada risiko operasional, finansial, atau perubahan budaya organisasi yang terlewat.
- Perbarui analisismu jika diperlukan.

2

Identification of Weaknesses

2 Peninjauan Ulang: Apakah Ada Risiko yang Terlewat?

Mari kita evaluasi apakah kategori operasional, finansial, dan perubahan budaya sudah lengkap.

A. Risiko Operasional yang Mungkin Terlewat

- Gangguan Operasional Saat Go-Live**
 - Potensi downtime produksi atau transaksi.
 - Perlu strategi parallel run atau soft launch.
- Ketergantungan pada Vendor**
 - Risiko jika vendor lambat merespons atau gagal deliver.
 - Perlu SLA yang jelas & knowledge transfer.
- Keamanan Data & Cybersecurity**
 - ERP menyimpan data finansial & sensitif.
 - Perlu penetration testing & role-based access control.

☑ Risiko keamanan sebelumnya belum disebut secara eksplisit — ini perlu ditambahkan.

B. Risiko Finansial Tambahan

- Biaya Tersembunyi (Hidden Cost)**
 - Upgrade hardware.
 - Customization tambahan.
 - Biaya training tambahan.
- Cash Flow Disruption**
 - Jika sistem akuntansi terganggu saat transisi.

☑ Perlu diperluas dalam mitigasi finansial.

C. Risiko Perubahan Budaya Organisasi

- Perubahan Pola Kerja**
 - ERP meningkatkan transparansi → bisa memicu konflik internal.
 - Perubahan power structure antar departemen.
 - Kehilangan Knowledge Informal**
 - Proses lama yang undocumented hilang saat sistemisasi.
- ☑ Risiko budaya organisasi sebelumnya belum dibahas secara mendalam.

3

Revision

3 Analisis yang Diperbarui (Versi Lebih Komprehensif)

Setelah review, berikut tambahan risiko penting:

✓ Risiko Keamanan & Kepatuhan

- Data breach
 - Ketidaksihuan regulasi (misalnya pajak, audit trail)
- Mitigasi: audit IT, penetration test, compliance check sebelum go-live.

✓ Risiko Operasional Transisi

- Penurunan produktivitas 1-3 bulan awal.
- Mitigasi: parallel run + hypercare period.

✓ Risiko Budaya & Politik Internal

- Perlawanan dari middle management yang kehilangan kontrol.
- Mitigasi: komunikasi manfaat berbasis peran + insentif adopsi.

✓ Risiko Knowledge Gap

- Ketergantungan pada konsultan.
- Mitigasi: dokumentasi lengkap + transfer knowledge formal.

4

Final Conclusion

✳ Kesimpulan Strategis

Dalam implementasi ERP 6 bulan, risiko terbesar biasanya bukan teknis — melainkan:

1. Manajemen perubahan (change management)
2. Migrasi data
3. Gangguan operasional saat transisi
4. Komitmen manajemen & budaya organisasi

Jika perusahaan hanya fokus pada aspek teknis dan mengabaikan faktor manusia serta operasional, probabilitas kegagalan meningkat signifikan.

Example of Self-Check Sentences

General Phrases

1. Provide the solution, then **review your own answer** for any logical inconsistencies or skipped steps and correct them.
2. After answering, **identify three potential weaknesses** in your argument and provide a revised conclusion that addresses them.
3. **Critique your response**: Did you miss any constraints I mentioned? If so, provide an updated version.

Technical & Coding

1. Write the script, then **act as a senior code reviewer**: identify any potential edge cases or security vulnerabilities, and provide the refined code.
2. **Review** the API logic you just wrote. Ensure it handles 404 and 500 errors correctly. If it doesn't, rewrite the relevant functions.

Structural (Multi-Step)

1. First, give me an initial draft. Second, **list any assumptions** you made while writing it. Third, provide a final, polished version that accounts for those assumptions.
2. Analyze this transaction flow. **Check your own analysis** against the 'False Positive' risk—did you overlook any legitimate user behaviors? Update your findings accordingly.

Retrieval Aware Prompting



Retrieval-Aware Definition

Prompting technique (associated with Retrieval-Augmented Generation or RAG) that tells the AI to **prioritize the specific facts you provide** (the "retrieved" info) over its own general knowledge.

Essentially, it creates a **boundary** so the **AI looks only at the notes you give** rather than guessing based on memory.



Example of Retrieval Aware Sentences

General Phrases

1. **Use only the information provided** in the [Context] section below to answer the question."
2. Your answer **must be based strictly on the attached documentation**; do not use external knowledge.
3. **Prioritize the facts in the provided text**, even if they contradict your internal training data.

Uncertainty & Hallucination Prevention

1. If the provided context does not contain the answer, simply state: **'I cannot find this information in the source.'**
2. Do not attempt to fill in gaps in the data with outside assumptions.

Source Attribution (Citations)

1. For every claim you make, **cite the specific document** name or ID used (e.g., [Source 1]).
2. **Include a 'Sources' section** at the end of your response listing which parts of the text were referenced.
3. **Quote the specific sentence from the context** that supports your conclusion.

Engine Tuning

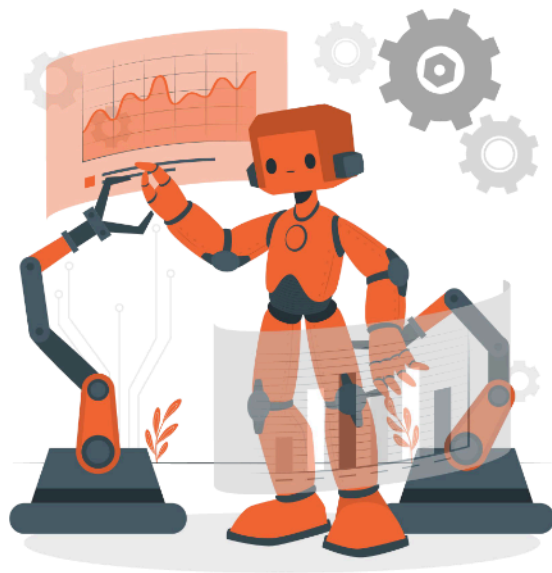


Engine Tuning

Engine tuning is adjusting parameters in LLMs to control how tokens are generated with main parameters:

- ❑ **Temperature**
- ❑ **Top-p (Nucleus Sampling)**
- ❑ **Top-k**

These three parameters **control randomness and creativity level**. They are only adjust how the model selects the next token.





Temperature, Top-P, Top-K Tuning

Temperature

Temperature controls how *random* or *creative* the model's output is. Temperature range is from 0 to 1.

- Low temperature (e.g., 0.1–0.3):
Output is more deterministic, focused, and predictable
- High temperature (e.g., 0.7–1.0+):
Output is more diverse, creative, and sometimes unpredictable



Top-P

Top-P limits the model to choosing from the smallest set of words whose cumulative probability $\geq P$. Top-P range is from 0 to 1.

Example:

If Top-P = 0.9 $\rightarrow$ model selects from the top words that together make up 90% probability mass

Effect:

- Keeps output coherent but still allows diversity
- Dynamically adjusts candidate pool size



Top-K

Top-K restricts the model to choosing from the top K most probable next words. Top-P range is from 0 to 100+.

Example:

If Top-K = 50 → model only considers the 50 highest-probability tokens

Effect:

- Smaller K → more focused and safe
- Larger K → more diverse but potentially noisy



Standard Operating Ranges

Temperature

- ❑ 0-0.3 → deterministic, focused, safe
- ❑ 0.5-0.8 → balanced
- ❑ 1.0-1.5+ → creative, risky, unpredictable

Top-P

- ❑ 0.9 → consider tokens covering 90% probability mass
- ❑ 0.5 → only very high probability tokens allowed

Top-K

- ❑ 1 → greedy (like temperature = 0)
- ❑ 5 → only top 5 tokens allowed
- ❑ 50 → more diversity

How the Three Parameters Interact

Most systems use: **Temperature + Top-p**

With typical setup:

- ❑ temperature = 0.7
- ❑ top_p = 0.9 or 1.0
- ❑ top_k = disabled

Why?

- Temperature controls creativity level
- Top-p prevents low-probability weird tokens
- Top-k becomes redundant if Top-p is used

Strategic Engine Tuning by Use Case

Use Case	Temperature	Top-P	Why
Financial reporting	0.1–0.3	0.8–0.9	Stability
Code generation	0.2–0.5	0.9	Reduce weird outputs
Chatbot assistant	0.6–0.8	0.9–1.0	Natural conversation
Brainstorming ideas	0.9–1.2	0.95–1.0	Encourage exploration



Managing Creativity vs Accuracy

Creativity vs Accuracy

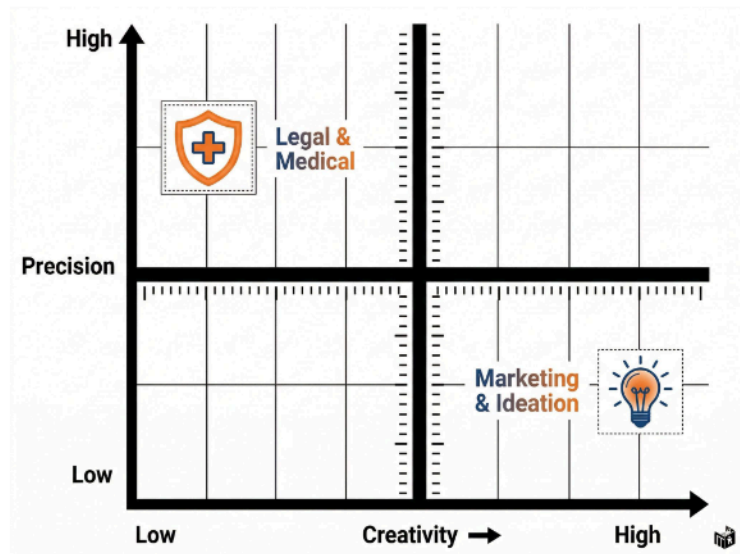
The graph illustrates the trade-off between Creativity (horizontal axis) and Precision/Accuracy (vertical axis) in Large Language Model (LLM) outputs.

Vertical Axis (Precision)

- ❑ Bottom → Low precision
- ❑ Top → High precision

Horizontal Axis (Creativity)

- ❑ Left → Low creativity
- ❑ Right → High creativity



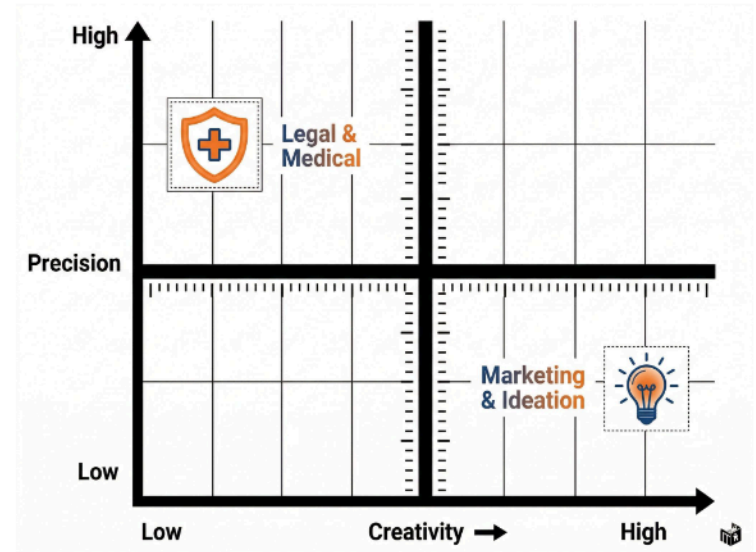
Creativity vs Accuracy (cont.)

The diagram shows that different use cases require different balances between creativity and precision.

Key Insight from the Graph:

As **creativity increases**, **precision tends to decrease** — but this is a trade-off tendency, not an absolute rule.

- ❑ **Higher randomness (higher temperature) encourages exploration**, but may reduce factual stability.
- ❑ **Lower randomness increases consistency and reliability.**



Context Window Management



Context Window Definition

In the world of Large Language Models (LLMs), the context window is the "**short-term memory**" of the model. It represents **the maximum number of tokens** (words, characters, or chunks) the model can process in a single interaction.



References:

<https://www.ibm.com/think/topics/context-window>

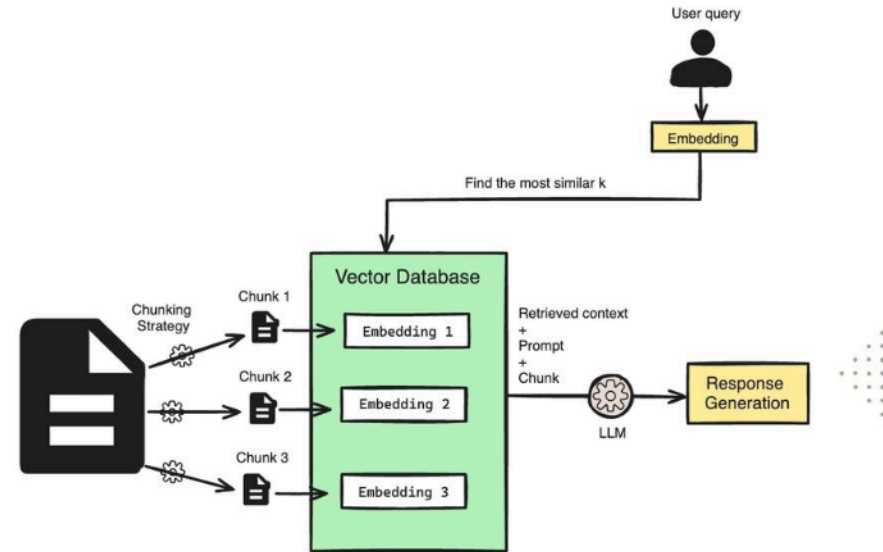
<https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-a-context-window>



Chunking

Chunking Definition

- ❑ **Chunking** is the **process of splitting large text or documents into smaller, manageable pieces (chunks)** before sending them to an LLM.
- ❑ **Chunking** is **breaking large documents into smaller semantic units** so LLMs can process them efficiently and accurately.



Type of Chunking

Character-Based (Fixed-Size) Chunking

- Split the text into a set number of characters, regardless of content or structure.
- **How it works:** define a `chunk_size` (e.g., 1000 characters) and a `chunk_overlap` (e.g., 200 characters). The overlap ensures that a sentence cut in half in one chunk is preserved in the next.
- **Example:**
 - Input: A 5,000-character transcript.
 - Result: 5 chunks of 1,000 characters each.
- **Best for:** Very consistent data where structure doesn't matter much.

Recursive Character Chunking

- **How it works:** It tries to split by a hierarchy of characters: first by double newlines (`\n\n`), then single newlines (`\n`), then spaces (), and finally individual characters.
- **Example:** If a paragraph is too large for the limit, it will split at the nearest sentence end rather than mid-word.
- **Best for:** Most text documents, as it keeps paragraphs and sentences together as much as possible.

Document-Specific (Structural) Chunking

- This respects the inherent format of the file (HTML, Python, or JSON).
- **How it works:** It splits based on logical headers (e.g., `# Header 1` in Markdown) or code blocks.
- **Example:**
 - Input: A Python script.
 - Result: One chunk per function or class definition.
- **Best for:** Technical documentation or code repositories where the "meaning" is tied to the structure.



Sliding Windows

- Initial Window**

Once	Upon	A	Time	There	Was	A	Kid	Who	Had	A	Dream
------	------	---	------	-------	-----	---	-----	-----	-----	---	-------

Active neurons in the initial window

Sliding Window

Once	Upon	A	Time	There	Was	A	Kid	Who	Had	A	Dream
------	------	---	------	-------	-----	---	-----	-----	-----	---	-------

Active neurons in the new window

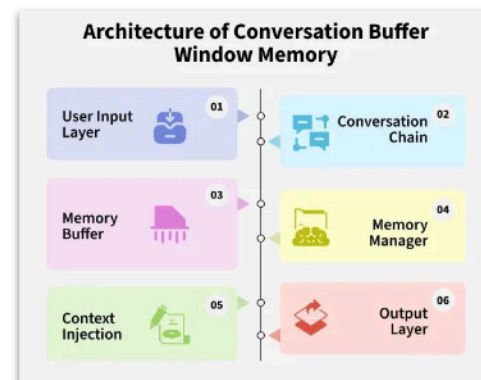
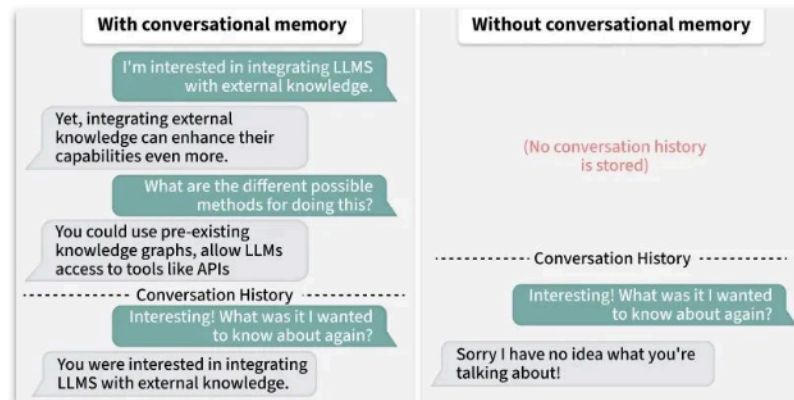
Neurons to be deleted
 Neurons from initial window
 New Neurons



Buffer Memory

Buffer Memory Definition

- Instead of deleting data, it use a **"compressor."**
- **The Process:** When the context reaches 80% capacity, you trigger a "hidden" LLM call: "Summarize the conversation so far, keeping key facts like the user's name, the project goal, and current errors."
- **The Example:**
 - **Original Context:** 3,000 tokens of back-and-forth troubleshooting.
 - **Managed Context:** A 200-token summary + the last 500 tokens of active chat.
- **Best for:** Customer support bots or creative writing where "plot points" must be preserved.



Cost Optimization Techniques



Cost & Performance Optimization

This table presents several prompt optimization techniques used in LLM systems to improve efficiency and control.

Technique	Definition	Cost Impact
Token Budgeting	Limiting the maximum output length.	-30% Cost
Prompt Compression	Removing unnecessary adjectives and filler words.	-15% Latency
System Message	Placing static instructions at the system level.	Consistency Up



Token Budgeting

Controlling Cost with Token Budgeting



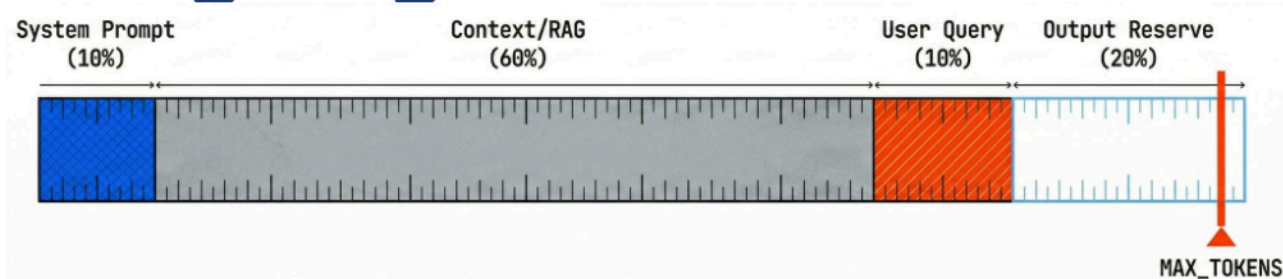
Token budgeting is **planning how many tokens are used for system instructions, retrieved context, user input, and model output** which control operational cost and ensure stable performance.

Why It's Important?

- ❑ LLM cost scales linearly with token usage. **More input + output tokens = higher cost.**
- ❑ Prevents runaway or excessively long outputs with **setting max_tokens** ensures responses stay within limits.
- ❑ Reduces latency with **fewer tokens → faster processing.**



Example of Token Budgeting



➤ Example allocations with the percentages (10%/60%/10%/20%) are not fixed rules.

➤ **Allocation depends on:**

- Model context window (e.g., 8K, 32K, 128K tokens)
- Use case (chatbot vs RAG vs summarization)
- Expected output length

➤ **For example:**

- RAG-heavy systems → Context may take 70%+
- Short Q&A → Context may take only 30–40%

➤ The red line labeled **MAX_TOKENS** indicates the total allowed output tokens must fit within the remaining context window. So, **max_tokens usually controls output tokens** only, not total context.

➤ The total context window = input tokens + output tokens combined.



Prompt Compression

Controlling Cost with Prompt Compression

The process of **reducing the number of tokens in a prompt** by **removing unnecessary words, redundant instructions, and filler content** while preserving the original meaning and intent.

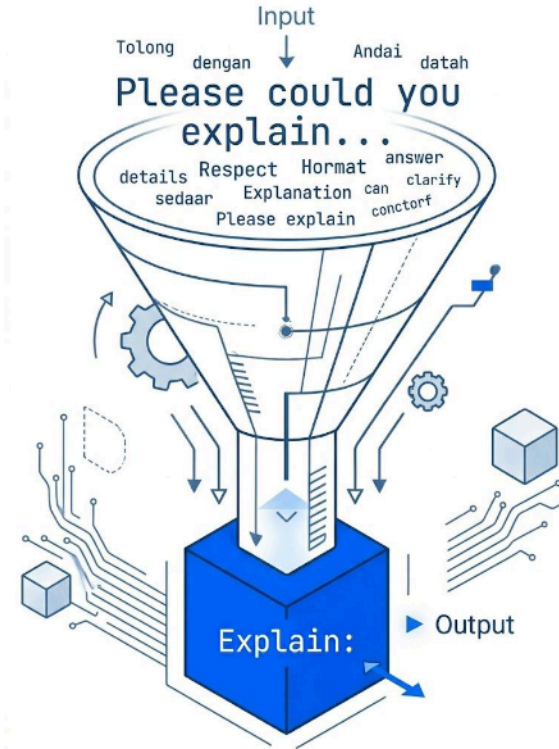
Example:

✗ Before

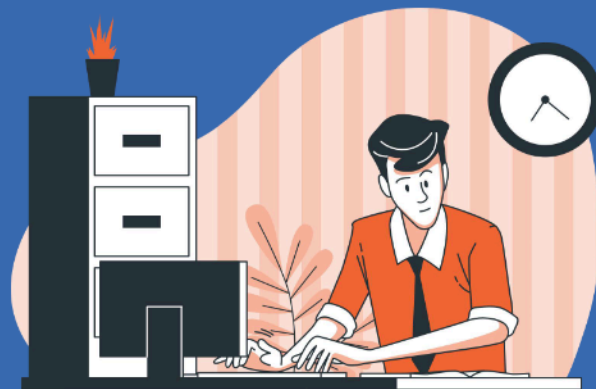
You are a professional financial analyst with 15 years of experience. Please carefully analyze the following data and provide a well-structured and properly formatted answer using bullet points. Make sure your explanation is clear and easy to understand.

✓ After

Analyze the following financial data and summarize key insights.



Guided Hands-on



Tuning LLM Parameters Practice

Learning Goals

By the end of this session, participants will be able to:

- Understand how Temperature, Top-P, and Top-K affect output.
- Control creativity vs consistency.
- Choose the right configuration for different use cases.

Setup

Participants can use:

- Google Colaboratory:
<https://drive.google.com/file/d/1IE7aWzsdYConEK4Zej1Y8-uETW9xB17r/view?usp=sharing>
- LLM API Key Gemini, GPT, or Claude

Base Prompts

Use this exact prompt for all experiments:

```
Explain why AI is important in daily life in 3-4 sentences.
```

Part 1: Temperature (Creativity)

Steps

1. Set Temperature = 0.0
2. Run → Save as *Output A*
3. Set Temperature = 0.5
4. Run → Save as *Output B*
5. Set Temperature = 1.0
6. Run → Save as *Output C*

Observe

- Is the answer always the same?
- Which output feels more "creative"?
- Which one is safest / most factual?

Key Insight

- 0.0 → deterministic (same answer, very safe)
- 1.0 → creative, diverse, less predictable

Part 2: Top-P (Probability Mass)

Steps

Fix Temperature = 0.7

- Top-P = 0.2 → Run → Output D
- Top-P = 0.5 → Run → Output E
- Top-P = 1.0 → Run → Output F

Observe

- Does vocabulary change significantly?
- Is output more limited or more expressive?

Key Insight

- Low Top-P → conservative
- High Top-P → more diverse choices

Part 3: Top-K (Token Limitation)

Steps

Fix Temperature = 0.7

1. Top-K = 10 → Output G
2. Top-K = 40 → Output H
3. Top-K = 100 → Output I

Observe

- Does the model feel "restricted"?
- Any unusual or creative wording?

Key Insight

- Low Top-K → predictable
- High Top-K → more variety

Part 4: Combine Parameters

 Run These Configurations

Config	Temperature	Top-P	Top-K
A	0.2	0.5	20
B	0.7	0.9	50
C	1.0	1.0	100

Chain of Thought Practice

Objective:

Compare reasoning quality with vs without CoT.

Scenario: Used Car Trade-In Pricing

Mitsubishi Xpander Ultimate 2019 A/T, KM 65.000, pajak hidup sampai Agustus 2026, ada lecet di bumper depan, service history lengkap di bengkel resmi. Harga pasaran sejenis Rp 195–215jt. Tentukan harga trade-in optimal.

Steps:

1. **Basic prompt:** *"Berapa harga trade-in mobil ini?"*
2. **CoT prompt:** *"Identifikasi faktor harga (tahun, KM, kondisi fisik, service history), bandingkan dengan harga pasaran, hitung depresiasi & adjustment kondisi, baru tentukan rekomendasi harga trade-in."*
3. Run both, paste outputs side-by-side.

Self-Check Practice

Objective:

Have AI critique and improve its own output

Scenario:

Customer test drove **Mitsubishi XForce Ultimate** yesterday but didn't commit. Draft a follow-up email.

Steps:

1. **Initial draft:** *"Tulis email follow-up untuk customer yang test drive Mitsubishi XForce Ultimate kemarin tapi belum membuat keputusan."*
2. **Self-Check layer:** *"Review email di atas. Identifikasi 3 kelemahan (terlalu pushy? kurang personal? CTA tidak jelas? tidak menonjolkan keunggulan XForce?). Tulis ulang versi yang lebih efektif untuk konversi."*
3. Compare initial vs revised.

Retrieval-Aware Practice

Objective:

Have AI critique and improve its own output.

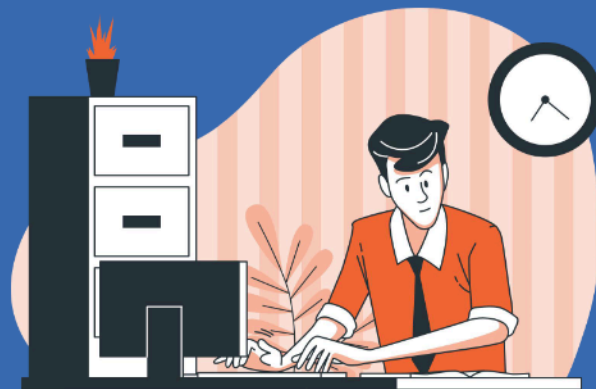
Scenario:

Customer test drove **Mitsubishi XForce Ultimate** yesterday but didn't commit. Draft a follow-up email.

Steps:

1. **Initial draft:** *"Tulis email follow-up untuk customer yang test drive Mitsubishi XForce Ultimate kemarin tapi belum membuat keputusan."*
2. **Self-Check layer:** *"Review email di atas. Identifikasi 3 kelemahan (terlalu pushy? kurang personal? CTA tidak jelas? tidak menonjolkan keunggulan XForce?). Tulis ulang versi yang lebih efektif untuk konversi."*
3. Compare initial vs revised.

Assignment



Prompt Optimization Assignment

Objective

Apply parameter tuning + reasoning technique + cost optimization to a Mitsubishi business use case.

Choose One Use Case

-  Sales pitch generator — pitch text for a specific Mitsubishi model
-  Lead nurturing email — follow-up to a test drive prospect
-  Marketplace listing — used Mitsubishi description for online listing
-  Customer service response — reply to delayed service complaint

Create Your Own Prompt

Open Google Colab:

https://drive.google.com/file/d/12nflvm_H6JWtIHJGD3YTKvMsNv2Prgc9/view?usp=sharing

Edit the variables:

```
YOUR_USE_CASE
= "Sales pitch generator"
YOUR_REASONING_TECHNIQUE =
"CoT"
YOUR_COST_STRATEGY
= "Token Budgeting"
YOUR_FINAL_PROMPT
= " " " " " " " "
```

Dummy Data

Inventory:

- Xpander Ultimate 2024, KM 8.000, Rp 295jt, 4 unit
- Xpander Cross Premium 2024, KM 5.000, Rp 335jt, 3 unit
- XForce Ultimate 2024, KM 3.500, Rp 405jt, 5 unit
- Pajero Sport Dakar 2024, KM 6.000, Rp 720jt, 2 unit
- Triton Athlete 4x4 2023, KM 18.000, Rp 525jt, 2 unit

Customer profile:

- Budi Santoso, 35 thn, keluarga 4 orang, budget Rp 280–340jt

Service center:

- Avg waktu servis 2–3 jam, ganti oli Rp 450k, tune-up Rp 950k

Run Experiments

 Run These Configurations

Experiment	Temperature	Top-P	Top-K
1 (Safe)	0.2	0.5	20
2 (Balanced)	0.6	0.8	50
3 (Creative)	1.0	1.0	100

Analyze Results

 Fill This Table

Criteria	Exp 1	Exp 2	Exp 3
Clarity			
Creativity			
Consistency			
Reasoning Quality			
Groundedness			
Token Efficiency			
Best Output			



Thank You

